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Adequate Prompting Improves Performance of Regression Models of Emotional Content

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ABSTRACT

Regression models of natural language content are widely popular in many applications, such as sentiment analysis, stance detection and emotion detection. The multidimensional Russell and Mehrabian's Valence-Arousal-Dominance (VAD) model has been universally recognized as a valuable tool for describing emotional content. Standard methods for VAD prediction learn the relation between natural language text and VAD values by fine-tuning a pre-trained machine learning model based on Transformers (e.g. BERT) on a corpus of natural language sentences manually annotated with VAD. We investigate the potential of employing effective prompting, a technique previously proven to be advantageous in classification and other natural language processing (NLP) tasks, to enhance the VAD prediction process. Our findings reveal that with appropriate prompting, we can leverage the knowledge acquired during pre-training to improve regression performance, showcasing the benefits of this approach for VAD prediction.

CCS CONCEPTS

• **Information systems → Data mining**; • **Computing methodologies → Supervised learning by regression**.

KEYWORDS

NLI, VAD detection, Emotion detection, NLP

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1 INTRODUCTION

The field of Natural Language Processing (NLP) has undergone significant transformations in recent years, largely fueled by the advent of large-scale language models. Among the various strategies developed to maximize their effectiveness, prompting has emerged as a compelling technique. It has been extensively applied for classification, sequence generation, and multiple other NLP tasks with noteworthy success [15].

Despite the widely acknowledged utility of prompting in a multitude of tasks, the nuances of employing for regression have been understudied. This paper extends the current understanding of prompting exploring the potential benefits of using prompts in a regression task, where the target variable is continuous. In particular, we focus on the well-established Russell and Mehrabian's Valence-Arousal-Dominance (VAD) model [23] of emotions and endeavor to predict the continuous values of Valence, Arousal, and Dominance associated with a natural language text.

The VAD model is widely used in psychology and related fields as it provides a comprehensive approach to identifying, understanding, and measuring emotional responses. The model's three dimensions are thought to capture the full range of human emotions. The VAD model describes and measures emotional states based on three orthogonal dimensions: Valence (V), Arousal (A) and Dominance (D).

- Valence: the quality of the emotion in a spectrum from negative to positive feelings. Negative valence includes emotions like sadness, anger, and fear, whereas positive valence can include joy, love, and excitement. It's a measure of the pleasantness or unpleasantness of an emotional state.
- Arousal: the intensity or energy associated with the emotion. It ranges from feelings of low arousal such as calmness, boredom, or fatigue, to feelings of high arousal like excitement, stress, or anxiety. It represents the level of activation or stimulation in an emotional state.
- Dominance: the perception of control or influence within the emotional state. Low dominance can indicate feelings of lack of control, submission, or being overwhelmed, while high dominance corresponds to feelings of control, empowerment, or influence.

In this paper we show that adequate prompting can be beneficial for the VAD regression task and propose two novel methods for VAD prediction that outperform existing approaches.

A baseline for VAD prediction consists on fine-tuning a pre-trained language model based on Transformers (e.g. BERT or RoBERTa) on a corpus of natural language sentences manually annotated with VAD values (e.g. EmoBank [1]). To date the top performing method is the approach of Park et al., which involves classification models trained on SemEval [18], ISEAR [24], and GE [5] datasets and then fine-tuned on EmoBank. This approach is appealing since it leverages the power of transfer learning, where models trained on

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an emotion classification task are further fine-tuned on the VAD prediction task. However, it requires explicit knowledge on emotion detection, provided by manually annotated corpora.

We show that it is possible to obtain competitive performances by simply adding suitable prompts to a pretrained (mask-based) model or an NLI-based model and fine-tuning on the VAD annotated dataset. In contrast to the Park et al. [20] approach, our method does not require pre-training on other annotated emotion detection datasets but is able to better exploit the knowledge of the emotional domain that the model has acquired during pre-training.

We evaluate our approach on two benchmarks in English [1] and Italian [7], respectively, and show that it outperforms all competitors. Although it does not rely on other emotion detection datasets for training, our approach outperforms the Park et al. method, showing that the emotional knowledge embedded in pre-trained models can be successfully exploited for improving the VAD prediction task. Our results suggest that adequate prompting might be beneficial to general regression tasks. Although prompting has been widely investigated in various NLP tasks, we are not aware of any approach that employs it for regression.

The contribution of our work can be summarized as follows:

- we investigate prompting on a widely known regression task (VAD prediction) and show that it improves results;
- we propose two novel methods for VAD prediction and show that they achieve state-of-the-art performances even without taking advantage of other specific emotion-annotated datasets (as the current state of the art [20] does);
- we perform an extensive evaluation on two different languages (English and Italian) and show that our findings generalize across languages.

The remainder of this paper is organized as follows: Section 2 discusses past work related to ours, Section 3 describes the corpora employed for our evaluation, Section 4 presents our approach, Section 5 shows the results of our experimental analysis and Section 6 discusses the findings of our work. Eventually, Section 7 concludes the paper and highlight future research directions.

2 RELATED WORKS

Emotion models in both psychology and natural language processing (NLP) can be broadly classified into two categories: categorical and dimensional. Categorical systems represent emotions as distinct and discrete states, such as the Basic Emotions proposed by Ekman [6] or the Wheel of Emotion introduced by Plutchik [21]. In contrast, dimensional models describe emotional states using a small set of independent dimensions. Among these dimensional models, the VAD model is particularly notable, as it defines emotional profiles based on Valence, Arousal, and Dominance, which are represented as real-valued vectors in a three-dimensional space. While categorical models were widely used in early NLP research, dimensional models have garnered increasing attention in recent years, especially in tasks involving emotion prediction for words and documents [1, 14, 29, 32].

Recently, much of the research in emotion analysis has focused on developing models that specifically target two dimensions of the VAD model: valence and arousal (VA). For instance, Cheng et al. [3] introduced a Bi-directional Long Short-Term Memory (BiLSTM) model that incorporates VA values for sentiment analysis in government social management. The winning team of Task 1 in the SemEval-2018 challenge [18] proposed a unified architecture, leveraging an ensemble of prediction models and diverse feature extraction methods, to tackle both subtasks.

To address subjective variations in emotional perception, VA-based models have been extended to represent emotions as probability distributions in the VA emotion space. Gaussian parameter-based approaches, commonly used in music and speech emotion analysis, have been employed to estimate emotion distributions by considering covariance information [33]. Noteworthy contributions in this area include works by Yang et al. [31], Zhao et al. [34], Sun et al. [27], Pribil et al. [22], Huang et al. [11], Trabelsi et al. [28], and Ghafoor et al. [8], which propose context-aware emotion classifiers utilizing Gaussian Mixture Models (GMMs) in the VA space. These models capture nuanced embedded emotions and variable perceptions in textual data.

Mendes et al. [17] present a significant advancement in estimating VA values by introducing a multilingual model that predicts real-valued emotional evaluations from text instances. They train an XLM-RoBERTa model [4] on a multilingual emotional corpus, demonstrating an interesting multilingual approach. However, these studies have overlooked the crucial dimension of "dominance", which plays a vital role in enriching the description of the emotional profile by quantifying the degree of control on an emotional scale.

A number of studies have delved into the integration of machine learning techniques with EGG signal data to explore the three-dimensional perspective of the VAD model [9, 19, 25]. These works provide valuable insights into the psychological aspect of the VAD model. However, it is important to note that they do not specifically address the isolated aspect of natural language processing (NLP).

Among the studies in this domain, the ones most relevant to our work are those that integrate machine learning techniques for text-level VAD analysis. Notably, Buechel et al. [2] introduce a comprehensive dataset comprising 10,548 annotated items, specifically designed to align with the VAD model (for more information, refer to Section 3). Park et al. [20] introduce an advanced system for the aforementioned dataset. The proposed model offers accurate predictions of nuanced emotions by considering the continuous dimensions of VAD. It is trained using the Earth Mover's Distance (EMD) loss [10], which minimizes the discrepancy between the predicted VAD score distribution and the categorical emotion distributions aligned along the VAD continuum. Remarkably, this model can simultaneously classify emotion categories and provide VAD scores for a given sentence. The training process consists of two phases. In the first phase, the model is trained using the EMD loss to minimize the difference between the predicted VAD score distribution and the categorical emotion distributions along the VAD continuum. The authors refine the model for VAD prediction using three distinct corpora with categorical labels, namely SemEval, ISEAR, and GoEmotion. The second phase of training focuses on the EmoBank dataset. By leveraging the model on the EmoBank dataset, the authors establish it as the new state-of-the-art for this dataset, effectively improving its performance. Despite the method's innovative approach, the system proposed by Park et al. relies on training with multiple corpora that contain categorical emotion annotations.

This dependence on categorical labels may restrict the system's capability to fully capture the intricate complexity and variability of emotional experiences, as these labels often oversimplify the underlying emotional states. Additionally, the obtained results from the three emotion datasets utilized in the study are not consistent, as the optimal values for valence, arousal, and dominance cannot be attributed to a single configuration of the model.

3 MATERIALS

The EmoBank corpus [2] represents a comprehensive and meticulously annotated dataset consisting of 10,548 English sentences. The annotations are based on dimensional emotion metadata, utilizing the VAD representation format. The development of this dataset strictly adheres to specific and well-defined criteria, which govern the data selection process. To ensure a comprehensive representation of language use, the EmoBank corpus encompasses multiple genres and domains of General English. It is constructed by incorporating several categories sourced from the Manually Annotated Sub-Corpus of the American National Corpus (MASC) [12, 13] and the corpus of SemEval-2007 Task 14 Affective Text (SE07) [26]. MASC provides annotations at various linguistic levels, while SE07 contributes annotations based on Ekman's six basic emotions using a [0; 100] scale. The annotation process includes a pilot study aimed at assessing the inter-annotator agreement (IAA) among different annotation perspectives, specifically those of the writer and the reader, across diverse domains. This comparative analysis involves two samples: one comprising movie reviews and the other drawn from a genre-balanced corpus. Based on the study findings, the dataset is annotated using a biperspective approach, considering both the writer's perceived emotion and the reader's perspective. The final dataset represents a weighted average of these dual perspectives. Recognizing the challenges posed by diverse emotion representation schemes in previous studies, a categorical emotion model was combined with the VAD dimensional ratings for annotation purposes. This additional annotation strategy enhances the comparability of results across different systems and promotes the reusability of the dataset. The data collection was facilitated through Amazon Mechanical Turk (AMT) and subsequently subjected to a manual verification process by domain experts. To address the writer's and reader's perspectives independently, separate annotation tasks were launched, ensuring a comprehensive and well-rounded annotation of the dataset.

The EmoITA dataset, developed by Gafà et al. [7] and made available as part of the EVALITA 2023 competition [1], represents a valuable resource for emotion analysis in the Italian language. This dataset comprises a total of 10,062 items. The EmoITA dataset is derived from the English version of EmoBank, which has undergone a manual translation and adaptation process to adapt to the Italian language. The structure of the corpus recalls that of the EmoBank dataset where each item is associated with a value for each VAD dimension. It is important to note that while the EmoITA dataset is based on the original version of EmoBank, some changes have been made to the value dimensions through manual reviews conducted by domain experts. This review process aims to ensure the adequacy and alignment of the data set with the nuances of the Italian language and cultural context. By leveraging the expertise of domain experts and incorporating their insights, the EmoITA dataset presents an improved and refined representation of emotional values within the Italian language.

4 METHODOLOGY

We propose a text-based regression approach to predict Valence, Arousal, and Dominance scores. To achieve this, we leverage pre-trained RoBERTa-based models obtained from the HuggingFace library. Our approach involves performing multidimensional VAD regression tasks, leveraging prompting techniques for both mask-based models and linguistic inference-based models. In Section 4.1, we provide a comprehensive overview of our method, namely Prompting-based Emotion Regression from Transformers (PERT), which is based on the RoBERTa-large architecture. Then, in Section 4.2, we present PERT-NLI, a framework for the regression task that explores the potential of prompting in linguistic inference systems.

4.1 PERT

To identify the values of Valence, Arousal, and Dominance in English and Italian short text, we use pre-trained linguistic models based on the RoBERTa architecture. For the English language, we employ the RoBERTa-large model [2], while for the Italian language, we use XLM-RoBERTa-large model [3]. The RoBERTa architecture [16] builds upon BERT with a different pretraining scheme. Notably, it removes the pre-training objective of predicting the next sentence and instead trains with mini-batches and larger learning rates. Unlike the BERT architecture, RoBERTa employs a dynamic masking mechanism during pre-training. This means that tokens are masked differently in each epoch, whereas BERT performs masking once and for all. The XLM-RoBERTa architecture [4] employed for the Italian task differs from RoBERTa-large in that it is trained on 2.5TB of filtered CommonCrawl data, enabling it to handle a multi-language task involving up to 100 different languages. To detect VAD values from text, we refined models based on RoBERTa using labeled data from the EmoBank dataset in both English and Italian. To fit the models to the multidimensional regression task, we incorporate a dropout and final linear layer with an output dimension of three (to recall the three values of valence, arousal, and dominance).

During the training phase, we incorporate different types of prompts to assist the model in identifying the regression task. This does not add any computational complexity to the training phase, as the model used for training remains unchanged. Thus, the introduction of new prompt types does not significantly affect the average running or training time, ensuring efficient model refinement without additional overhead. We utilize three distinct prompting strategies in our experiments. The first strategy, Prompt 1 (P1), aligns with the nature of the model by employing a masking task. It is defined as follows: "Context: [Input Text] - Valence: The emotional valence of the context is <mask>. - Arousal: The level of

[1] https://www.evalita.it/campaigns/evalita-2023/

[2] https://huggingface.co/roberta-large

[3] https://huggingface.co/xlm-roberta-large

arousal in the context is <mask>. - Dominance: The perceived dominance associated with the context is <mask>. Please predict the missing value for each dimension using the context provided." This prompt contextualizes the context and guides the model to replace the masked tokens with appropriate values, leveraging the power of the masking task. The second strategy, Prompt 2 (P2), contextualizes the model without explicitly mentioning the <mask> token. It is defined as: "Context: [Input Text] - Valence: What is the emotional value of the context? - Arousal: What is the level of arousal in the context? - Dominance: What is the perceived dominance associated with the context? Please answer the questions for each dimension using the context provided." In this case, we evaluate the model's performance solely based on contextualization, without relying on explicit masking. The third strategy, Prompt 3 (P3), involves providing only keywords without detailed explanations. We define Prompt 3 as the sentence "[Input Text] - Value, arousal, dominance -". This prompt aims to explore the extent to which the detailed context explanation affects the model's performance by providing minimal information through keywords. For the Italian language, we apply the same types of prompts translated into the target language.

Both the RoBERTa and XLM-RoBERTa -based models are fine-tuned with a learning rate of $1e-5$, a batch size of 64, a dropout of 0.1, and AdamW as the optimizer. The Mean Squared Error (MSE) is adopted as the loss function [4]. MSE is a commonly used metric for evaluating the performance of regression models. It measures the average squared difference between the predicted values and the actual values of a target variable. MSE is calculated as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where n refers to the number of data points or observations, y_i represents the actual values of the target variable and $\hat{y}_i$ indicates the predicted values of the target variable.

The described fine-tuning process enhances the models by specializing them in the detection of VAD values. This is combined with the use of prompting techniques, which further improve their performance. We refer with "PERT" to the model for English and with "XML-PERT" for Italian. Figure 1 shows an overview of the PERT model.

4.2 PERT-NLI

We introduce the PERT-NLI and XML-PERT-NLI models, based on the MNLI-RoBERTa-large [5] and MNLI-XLM-RoBERTa-large [6] systems designed to solve natural language inference tasks on the MNLI common sense dataset [30]. While the MNLI-RoBERTa model focuses on the English language, the MNLI-XLM-RoBERTa system is a multilingual model fine-tuned on an automatically-translated Italian version of the MNLI dataset. To adapt the systems for the VAD multidimensional regression task, we conduct fine-tuning of these models on the EmoBank dataset, considering both English and Italian languages. During training, we replace the last linear layer of the models with a modified layer tailored for the VAD multidimensional regression task, and we incorporate a dropout layer and an additional linear layer. Our NLI-based systems employ input sentences as premises and pair them with predefined fixed prompts acting as hypotheses. These prompts play a crucial role in guiding the model training process, focusing on specific assignments.

In line with the approach taken by the PERT and XML-PERT models described in Sect. 4.1, we employ three different prompt configurations. While the second and third configurations maintain the same structure as previously described, the first configuration aligns more closely with the model's nature. It utilizes declarative sentences that aim to direct the model's attention toward a specific level of implication. To achieve this, we have developed the following prompt: "Context: [Input Text] - Valence: the emotional valence of the context is high. - Arousal: the level of arousal in the context is high. - Dominance: the perceived dominance associated with the context is high.". This prompt guides the model to confirm the definitions related to each dimension and assign high regression scores when the statement aligns with the truth, and lower scores when the regression values deviate from the truthfulness of the statement.

For the Italian version, a translated hypothesis based on the English prompt is utilized. The PERT-NLI models are trained with a learning rate of $1e-5$, a batch size of 64, a dropout of 0.1, and AdamW as the optimizer. Again, the Mean Squared Error (MSE) is adopted as the loss function.

5 RESULTS AND EVALUATION

We perform an extensive experimental analysis to evaluate the effectiveness of our approach and compare it against baseline and state-of-the-art methods for VAD prediction in both English and Italian. In Sect. 5.1, we present the outcomes of our dimensional emotion analysis, assessing the performance of our methods on the English version of the EmoBank dataset. Then, in Sect. 5.2, we evaluate the results achieved on the EmoITA dataset.

5.1 English multidimensional analysis

We evaluate the performance of our models in the VAD multidimensional regression analysis using the EmoBank dataset. We compare the results of our PERT and NLI-PERT models, which were developed based on the method described in Section 4, considering all the model configurations and prompt variants. To assess the effectiveness of our approaches, we compare our regression models with the Park et al. approach [20] and consider their three different models trained on SemEval, ISEAR, and GoEmotion, respectively, and subsequently refined on the EmoBank dataset. Table 1 summarizes the approaches considered in our comparison, highlighting the data they have been fine-tuned on. Mnli-Roberta and PERT-NLI have been subjected to a first fine-tuning step on MNLI, while Park et al. methods have employed datasets for Emotion classification. Note that our method does not use emotion-annotated datasets other than EmoBank for training.

Table 2 reports the Pearson correlation with the ground truth obtained by our models, namely PERT and NLI-PERT, alongside by Park et al.'s models. Additionally, we present the baseline outcomes achieved by fine-tuned RoBERTa-large and MNLI-RoBERTa-large,

[4]https://pytorch.org/docs/stable/generated/torch.nn.MSELoss.html
[5]https://huggingface.co/roberta-large-mnli
[6]https://huggingface.co/Jiva/xlm-roberta-large-it-mnli

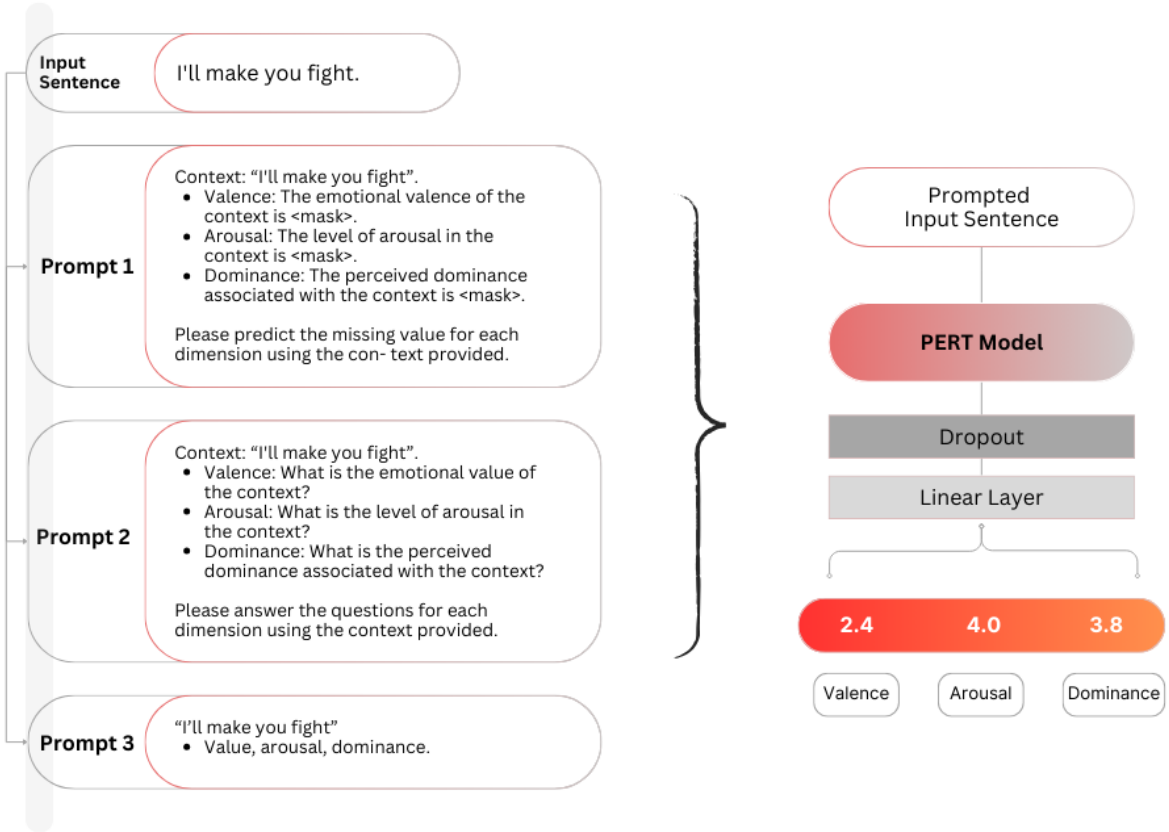

Figure 1: Overview of the PERT model. The prompt configurations (left) utilize three distinct prompting strategies to preprocess the input text. The modified text is then used to train the model (right), which consists of a RoBERTa-based architecture followed by a dropout layer and a final linear layer with three outputs, corresponding to the VAD regression targets.

which served as the foundation for our PERT and NLI-PERT models and do not employ any prompting techniques. For our models, we report the average of 10 runs for each prompt, along with the standard deviation. For Park et al. we report the correlation values declared by the authors [20], which refer to a single run.

The results demonstrate that the PERT models consistently outperform both the RoBERTa and Park et al.'s models in terms of correlation for valence, arousal, and dominance. Among the PERT models, it is evident that the first prompt configuration P1 achieved the highest correlation values for valence (0.844), arousal (0.587), and dominance (0.539), respectively. Additionally, the PERT-NLI method, which incorporates linguistic inference-based prompting, achieved competitive correlation values, albeit slightly lower than PERT. Overall, the results indicate that both PERT and PERT-NLI outperform the reference models RoBERTa and Mnli-RoBERTa as well as the models of Park et al.

5.2 Italian multidimensional analysis

We conducted an evaluation of our models for VAD multidimensional regression in Italian using the EmoITA dataset. In this section, we compare the results of the XML-PERT and NLI-XML-PERT models, as described in Sect. 4.2, with the baseline models trained on the same dataset without any prompt implementation, namely the XLM-RoBERTa models and MNLI-RoBERTa. While the XLM-RoBERTa model is multilingual and trained in Italian among other languages, the MNLI-RoBERTa model specifically focuses on the MNLI task in Italian. Table 3 presents the average Pearson correlation obtained by 10 runs for each configuration of the proposed methods, accompanied by standard deviation, for the Italian language. We observe that both XLM-RoBERTa and Mnli-XLM-RoBERTa performed well, achieving correlations ranging from 0.820 to 0.836 for valence, 0.669 to 0.679 for arousal, and 0.674 to 0.683 for dominance. Shifting our focus to the proposed models, the XML-PERT (Ours)

Table 1: Fine-tuning data employed by the approaches compared in this paper. All methods are based on the pre-trained LM RoBERTa-large. Mnli-RoBERTa and PERT-NLI are first fine-tuned on MNLI (a translated MNLI for the Italian version), then fine-tuned on EmoBank. The Park et al. methods are first fine-tuned on annotated datasets for emotion classification, then fine-tuned on EmoBank.

Model	NLI-based fine-tuning	Emotion classification fine-tuning	VAD prediction fine-tuning
RoBERTa	-	-	EmoBank
Mnli-RoBERTa	MNLI	-	EmoBank
(Park et al.), EB←SemEval	-	SemEval	EmoBank
(Park et al.), EB←ISEAR	-	ISEAR	EmoBank
(Park et al.), EB←GE	-	GoEmotion	EmoBank
PERT (Ours)	-	-	EmoBank
PERT-NLI (Ours)	MNLI	-	EmoBank

Table 2: Performance of PERT and NLI-PERT models in terms of average and standard deviation of Pearson correlation coefficient (r) scores obtained by 10 runs for valence, arousal, and dominance values on the EmoBank dataset. We present three different prompt variations for each model (P1, P2, P3) and compare our methods with the SOTA results reported by Park et al. (EB←SemEval, EB←ISEAR, EB←GE) as well as the baseline models (RoBERTa and Mnli-RoBERTa). For Park et al. we considered the correlation values declared by the authors [20], which refer to a single run.

Model	V (r)	A (r)	D (r)
RoBERTa	0.822 (± 0.004)	0.571 (± 0.005)	0.511 (± 0.008)
Mnli-RoBERTa	0.825 (± 0.005)	0.568 (± 0.014)	0.488 (± 0.026)
(Park et al.), EB←SemEval	0.838	0.570	0.518
(Park et al.), EB←ISEAR	0.836	0.568	0536
(Park et al.), EB←GE	0.835	0.573	0.529
PERT (Ours) - P1	**0.844** (± 0.005)	**0.587** (± 0.003)	**0.539** (± 0.006)
PERT (Ours) - P2	0.840 (± 0.004)	0.586 (± 0.005)	0.536 (± 0.011)
PERT (Ours) - P3	0.829 (± 0.004)	0.576 (± 0.005)	0.523 (± 0.010)
PERT-NLI (Ours) - P1	0.839 (± 0.002)	0.579 (± 0.005)	0.498 (± 0.021)
PERT-NLI (Ours) - P2	0.833 (± 0.003)	0.572 (± 0.009)	0.501 (± 0.024)
PERT-NLI (Ours) - P3	0.837 (± 0.002)	0.575 (± 0.005)	0.524 (± 0.007)

Table 3: Performances of XML-PERT and XML-NLI-PERT models (Italian) in terms of average Pearson correlation coefficient (r) scores obtained by 10 runs for valence, arousal, and dominance values on the EmoITA dataset. We present three different prompt variations for each model (P1, P2, P3) and compare our methods with the baseline models (XLM-RoBERTa and Mnli-XLM-RoBERTa). All results are accompanied by their corresponding standard deviation.

Model	V (r)	A (r)	D (r)
XLM-RoBERTa	0.820 (± 0.014)	0.669 (± 0.013)	0.674 (± 0.009)
XML-PERT (Ours) - P1	**0.834** (± 0.002)	**0.682** (± 0.003)	**0.682** (± 0.003)
XML-PERT (Ours) - P2	0.820 (± 0.009)	0.668 (± 0.013)	0.669 (± 0.009)
XML-PERT (Ours) - P3	0.826 (± 0.004)	0.675 (± 0.007)	0.676 (± 0.003)
Mnli-XLM-RoBERTa	0.836 (± 0.003)	0.679 (± 0.003)	0.683 (± 0.002)
XML-PERT-NLI (Ours) - P1	0.834 (± 0.003)	**0.683** (± 0.003)	0.682 (± 0.002)
XML-PERT-NLI (Ours) - P2	0.832 (± 0.003)	0.680 (± 0.004)	0.677 (± 0.002)
XML-PERT-NLI (Ours) - P3	**0.837** (± 0.003)	0.681 (± 0.003)	**0.685** (± 0.003)

and XML-PERT-NLI (Ours) models demonstrated comparable or slightly improved performance compared to the reference models. In particular, the XML-PERT model with the P1 configuration exhibited significant improvements over the relative baseline (XLM-RoBERTa), achieving a correlation of 0.834 for valence, 0.682 for arousal, and 0.682 for dominance. These results surpassed the baseline by 1-2 percentage points, with a difference significantly higher

than the standard deviation. Similarly, the XML-PERT-NLI models delivered competitive outcomes when compared to both PERT-XML and its respective baseline (MNLI-XLM-RoBERTa). In the P3 configuration, they achieved a correlation of 0.837 for valence, 0.681 for arousal, and 0.685 for dominance.

To summarize, our findings suggest that the proposed models, XML-PERT (Ours) and XML-PERT-NLI (Ours), exhibit improved performance in VAD regression for the Italian language compared to the baseline models. While the P1, P2, and P3 configurations produced similar or slightly different results, all demonstrated effectiveness in improving performances.

6 DISCUSSION

Our method demonstrates superior performance compared to existing approaches for the multidimensional VAD regression task in English and establishes itself as the new state-of-the-art for Italian as well. The advantage of the prompting method becomes even more pronounced when compared to the previous state-of-the-art approach in English. Unlike other methods, our approach leverages the formulation of appropriate prompts that adapt to the model architecture, effectively emphasizing the task context. Notably, our method achieves these results without requiring external data beyond the EmoBank (EmoITA for Italian) datasets.

Upon analyzing the results, we observe that prompts incorporating richer information (prompts 1 and 2) yield better performance in the English version. These prompts not only consider the sentence but also incorporate the contextual information of the task, leading to significant improvement. On the other hand, prompts closer to the baseline (such as P3), which merely add keywords, do not provide a substantial boost to the model's performance. In contrast, the first and second prompts are more self-explanatory and introduce a stronger semantic framework, resulting in improved performance. When examining the first prompt, we note that the PERT model performs better due to its familiarity with the prompt configuration, effectively leveraging its architecture (e.g., recalling the mask token). Conversely, the performance of the NLI models decreases despite showing comparable results. This decline can be attributed to the NLI models' training on a larger corpus of common-sense data that provides limited information for better contextualizing abstract concepts such as the emotional valence of a sentence or the degree of arousal and dominance associated with it. Consequently, their extensive training processes become a disadvantage when compared to PERT models, which are specifically designed and fine-tuned for the VAD regression task.

Shifting our focus to the results in Italian, we find that the XML-NLI-PERT models outperform or slightly surpass the XML-PERT models. This difference becomes more evident in the baselines, where the XML-NLI-Roberta base model outperforms XLM-RoBERTa by approximately one percentage point. This improvement can be attributed to the NLI model's training on a specialized MNLI dataset for the Italian language, which exposes it to more relevant Italian data. Interestingly, the proficiency of the model in VAD regression appears to benefit more from this exposure than from the contribution of the prompt itself. In this case, prompt 3 aligns closely with the baseline, recalling only the keywords of the task and surpassing the baseline by a small margin, demonstrating its effectiveness. This behavior is specific to NLI-based models, as observed from the comparison between the XML-PERT and XLM-RoBERTa models. The pattern aligns with the findings from the English version, where the presence of a more descriptive semantic prompt (P1) significantly aids the model in task execution.

7 CONCLUSION

In this paper, we explore the effectiveness of prompting in the context of VAD regression. By incorporating suitable prompts into pre-trained language models, we have demonstrated significant improvements in VAD prediction performance. Our approach surpasses the current state-of-the-art method by Park et al., which relies on explicit emotion detection datasets for pre-training. In contrast, our method leverages the knowledge embedded in general-purpose pre-trained models, eliminating the need for additional annotated datasets. This highlights the potential of pre-trained models in capturing emotional information and their applicability in the VAD prediction task. We have conducted extensive evaluations on two benchmark datasets in English and Italian, showcasing the generalizability of our approach across different languages. Our findings open up new possibilities for incorporating prompting techniques in regression tasks, expanding the repertoire of applications for pre-trained language models. Future research can explore further improvements in VAD regression and investigate the effectiveness of prompting in other regression domains.

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